Hierarchical Predictive Control of Network Traffic Signals Using Link Transmission Model With Queue Dynamics

Lei Wei, Konstantinos Ampountolas, *Member, IEEE*, Angelika Hirrle, and Meng Wang, *Senior Member, IEEE*

***Abstract*— Network signal control is an effective way to mitigate traffic congestion. However, most network signal control methods ignore the risk of queue spillback. Although local and decentralized control methods have the potential to address spillback, their performance at the system/network level is not guaranteed, making it challenging to achieve the global optimum. This study proposes a novel hierarchical model predictive control (MPC) approach that utilizes the link transmission model (LTM) with queue transmission at both road segment and turn levels to optimize traffic signals for road networks. At the network level, the controller employs a state-of-the-art LTM framework that can describe segment-level flow dynamics and uses quadratic programming to determine the effective fractions of green time for network throughput maximization. The network decisions of green time fractions are sent to the local layer as a reference. The local layer builds on a refined LTM with turn-level queue transmission and formulates a nonlinear programming problem to track the reference to ensure that the optimal network decision is realized at the individual intersections while eliminating potential spillback. Simulation experiments on both an arterial and a grid network are conducted to verify the performance of the proposed approach. The results reveal that the proposed MPC leads to substantial improvements in terms of throughput and queue length, especially in oversaturated conditions, and is robust against demand prediction errors.**

***Index Terms*— Model predictive control, traffic signal control, link transmission model, macroscopic traffic flow, queue spillback.**

I. Introduction

TRAFFIC congestion in large-scale urban road networks leads to excessive economic losses and emissions. Among many measures to deal with urban congestion, traffic signal control has been proven to be a cost-effective solution [1], [2]. However, signal control for urban road networks faces significant challenges due to the complex congestion phenomena, notably when spillback occurs [3]. Moreover, as the spatial or temporal scale of the network expands, signal control involves an exponentially growing number of state and decision variables. Searching optimal solutions for such high-dimensional problems is computationally prohibitive [4]. Therefore, the development of effective signal control schemes to mitigate traffic congestion in urban road networks is still much needed.

Received 16 August 2024; revised 17 February 2025 and 7 May 2025; accepted 7 May 2025. Date of publication 22 May 2025; date of current version 21 October 2025. The Associate Editor for this article was R. Malekian. *(Corresponding author: Meng Wang.)*

Lei Wei, Angelika Hirrle, and Meng Wang are with the Chair of Traffic Process Automation, Institute of Traffic Telematics, Technische Universität Dresden, 01069 Dresden, Germany (e-mail: lei.wei@tu-dresden.de; angelika.hirrle@tu-dresden.de; meng.wang@tu-dresden.de).

Konstantinos Ampountolas is with the Automatic Control and Autonomous Systems Laboratory, Department of Mechanical Engineering, University of Thessaly, 38334 Volos, Greece (e-mail: k.ampountolas@uth.gr).

Digital Object Identifier 10.1109/TITS.2025.3568869

A. Related Work

Among the network-wide traffic control, researchers and traffic engineers have explored various modeling and optimization methods, which can be broadly divided into three categories based on the control architecture: centralized, distributed, and hierarchical.

Centralized traffic control involves a control center that manages intersections across an entire network or a substantial portion of it. Benefits of centralized control in reducing congestion and externalities have been shown in literature, e.g., perimeter control using macroscopic fundamental diagrams (MFDs) [1], [5], store-and-forward modeling paradigm-based network-wide traffic signal control [6], [7], [8] and its extension to multi-commodity traffic signal control in the presence of connected vehicles [9]. However, centralized real-time traffic signal control is only applicable to relatively small-scale networks due to the complexity of the involved optimization problem. Moreover, large-scale road networks using centralized control require a high-level infrastructure investment and maintenance, and and is vulnerable to subsystem failures [10], [11]. Distributed control systems allow individual intersections to make independent control decisions based on the local conditions, e.g., max pressure or back pressure-based methods [12], [13], [14], but these methods face challenges in achieving network-wide optimization because limited coordination can be achieved among intersections. Hierarchical control offers a balance between centralized control's holistic view and coordination capabilities and distributed control's scalability, which has emerged as a promising choice for improving large-scale network performance. In this framework, the controlled systems are organized into hierarchies, with lower-level controllers making decisions for local intersections based on the local conditions and higher-level controllers coordinating and optimizing network traffic across intersections. The PRODYN [15] and SCATS [16] are typical bi-level approaches that have been successfully

1558-0016 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Nanyang Technological University Library. Downloaded on September 10,2026 at 05:12:33 UTC from IEEE Xplore. Restrictions apply.

implemented to coordinate the signals across the road network. Unfortunately, they are not suitable for oversaturated traffic conditions due to the rapidly changing flow operation and the false estimation or prediction of flow parameters (e.g., arrival rate) [6].

Traffic flow models have been proven important in designing and testing network signal control strategies, especially model-based strategies such as model predictive control (MPC) [8], [17], [18], [19], [20]. The MPC-based network control methods usually used macroscopic traffic flow models to formulate the network flow propagation, and many existing studies rely on the MFDs [1], [17]. The MFDs-based methods assume that the traffic state within a (sub-)network is homogeneous [21], [22], [23], [24], but this may not always hold for congested traffic states, especially under oversaturated conditions. Another group of studies used cell transmission models (CTMs) that divide a road segment (RS) into some spatial cells to formulate the flow dynamics, but the computational complexity caused by the sophisticated spatiotemporal discretization poses challenges in terms of scalability and real-time implementation in large networks [19], [25]. This is particularly the case in oversaturated conditions where we need to trace the queue development and spillback. Employing CTMs to trace queue will entail fine discretization of road segments, resulting in excessive computational overheads (more road cells involved in the optimization problem) [26], [27].

van de Weg et al. [19], [25] have used standard link transmission models (LTMs) as the prediction model for network traffic control. The standard LTM does not explicitly model turn-level queue dynamics within the road. Instead, it focuses on calculating aggregated inflows and outflows at the upstream and downstream boundaries of each link. Although the standard LTM can estimate vehicle accumulation within a link over time, this is not equivalent to queue formation. In congested conditions, accumulation includes both queued vehicles and those still moving upstream of the queue. The queue density typically varies between critical density and jam density [27]. Therefore, the standard LTM, which assumes only a constant jam density, is less capable of estimating the varying queue density and queue length within the link. Moreover, they treat traffic as a fluid flowing along the road, without distinguishing between different turning movements (e.g., left, right, or straight). Wei et al. [28] formulates a link-based flow model with turn-level queue transmission to simulate traffic flow propagation and queue dynamics. The queue length is crucial for measuring signal control performance [29]. However, their study only focused on model development and verification.

With the development of machine learning, intelligent control methods have gained considerable attention in signal control. Model-free deep learning/reinforcement learning-based approaches are used to train the agent for signal control scheme [30], [31], [32]. Zhou et al. [33] introduced a novel distributed deep reinforcement learning method for the network-wide signal control coordination, reducing communication overhead while ensuring control collaboration in reducing queue length. Ren et al. [34] introduces multi-agent coordinated policy optimization (MACoPO), a coordination algorithm based on multi-agent RL, for multi-intersection signal control. While these data-driven model-free signal control methods have shown the potential to improve traffic efficiency without the need for physical models, the interpretability of the controller is limited and safety constraints are difficult to address, raising concerns about trust and accountability in road environments. There is still a need to develop advanced model-based control methods.

B. Contributions

This study proposed a hierarchical MPC framework using LTM with queue transmission (MPC-Q) to optimize traffic signal for urban networks.The main contributions of this study are summarized as follows:

Firstly, this study extends the general link-based MPC framework to involve turn-level queue dynamics. Unlike existing link-based MPC methods that only consider aggregated flow dynamics at the segment level, the proposed MPC incorporates flow dynamics and queue information at both the segment and turn levels, enabling fine-grained signal control optimization.

Secondly, compared to existing link-based MPC methods, this study proposes a hierarchical control structure that allows modular design of the two layers at different spatial-temporal scales to adapt to real-time changes in turn-level queue lengths at local intersections, while keeping the property to seek system optimum over a long prediction horizon. This design enables the addition of turn-level queue transmission with computational tractability.

Specifically, an upper-layer network controller predicts network flow propagation with segment-level aggregated flow dynamics and determines the optimal total fraction of green time for all turning movements of an approach. The objectives of the network controller are to maximize the network throughput and minimize the oscillations in the signal plan. The optimal green time fraction of an approach is sent to a lower-layer local intersection controller as the reference trajectory. The local layer predicts the turn-level queue length dynamics for each turning movement in an approach and determines the turn-level optimal fraction of green time. The objective of the local layer is to minimize the tracking error and queue length. Experiments on the arterial and grid network are performed to demonstrate the performance of the proposed approach.

The rest of the paper is organized as follows. Section II-A provides the framework of the proposed MPC-Q. Then, Sections II-B and II-C formulate the proposed MPC-Q in detail, in which the two-layer controller design is discussed explicitly. Section III illustrates the numerical experiments in different scenarios and compares our proposed method with the existing methods to demonstrate the superiority of our method.

II. Formulation of the Hierarchical MPC Using LTM With Queue Dynamics

The MPC-Q proposed in this study employs the LTM with queue transmission to optimize traffic signals for urban networks. This section first presents the framework of the proposed MPC-Q. Then, it discusses the concept of the LTM

Authorized licensed use limited to: Nanyang Technological University Library. Downloaded on September 10,2026 at 05:12:33 UTC from IEEE Xplore. Restrictions apply.

TABLE I
NOMENCLATURE

Parameter	Definition	Parameter	Definition
b_i^{μ}	Effective fraction of green time in μ of i	$N_{i,\mu}^{que}(k)$	Cumulative queue inflow in μ of i at k
b_i^{seg}	Segment-level effective fraction of green time of i	$N_{i,\mu}^{out,des}(k)$	Desired cumulative outflow in μ of i at k
$\boldsymbol{b}$	Set of segment-level effective fractions of green time of all incoming road segments	$\boldsymbol{N}_i^{in}(k)$	Set of cumulative inflows in all turning directions of i at k
$\boldsymbol{b}_i$	Set of turn-level effective fractions of green time of i	$\boldsymbol{N}_{i,sh}^{in,max}(k)$	Set of maximum inflow with the shockwave traveling the queue backwards in all turning directions of i at k
$\boldsymbol{b}_{\mu}$	Set of turn-level effective fractions of green time of all incoming road segments	$\boldsymbol{N}_i^{out}(k)$	Set of cumulative outflows in all turning directions of i
$\Delta\boldsymbol{b}(k)$	Set of variation in effective fractions of green time at k	$\boldsymbol{N}_i^{out,des}(k)$	Set of desired cumulative outflow of i at k
$\boldsymbol{B}(k)$	Matrix related to effective fractions of green time at k	$\boldsymbol{N}_i^{out,max}(k)$	Set of cumulative outflow limit of i at k
$\boldsymbol{C}(k)$	Matrix related to demand at k	$\boldsymbol{N}_i^{que}(k)$	Set of cumulative queue inflows in all turning directions
Δt	Sampling time interval	$\boldsymbol{U}_k$	Control vector in prediction horizon
$d_{i,\mu}^{in}(k)$	Traffic demand in μ of i at k	$s_i(k)$	System state at k
$e_{i,\mu}$	Turning rate	$\boldsymbol{S}_k$	State vector in prediction horizon
$\boldsymbol{I}^c$	Set of conflict signal phases	$\boldsymbol{R}$	Weight matrix of control vector
i	Road segment	$\boldsymbol{Q}$	Weight matrix of state variable
$\boldsymbol{i}^O$	Set of origins directly upstream i	q_{cap}^{μ}	Origin capacity in μ
$\boldsymbol{i}^{up}$	Set of road segments directly upstream i	q_{cap}	Origin capacity at segment level
k	Time step	q_{μ}^{c}	Critical flow rate in μ of i
L_i	Length of road segment i	$q_{i,\mu}^{in}(k)$	Link inflow rate in μ of i at k
$L_{f,\mu}$	Free-flow part length in μ of i	$q_{i,\mu}^{out}(k)$	Link outflow rate in μ of i at k
$L_{q,\mu}$	Queuing part length in μ of i	$q_{i,\mu}^{que}(k)$	Queue inflow rate in μ of i at k
Π^O	Set of fictitious road segments at the origins	$q_{i,\mu}^{s}(k)$	Saturation rate in μ of i at k
Π^L	Set of general road segments	m	Number of road segments in the network
μ	Turning direction	ρ_{μ}^{c}	Critical density in μ of i
$n_{f,\mu}$	Constant of sampling instants related to $v_{f,\mu}$ and $L_{f,\mu}$	ρ_{μ}^{jam}	Jam density in μ of i
$n_{L,\mu}$	Constant of sampling instants related to $v_{f,\mu}$ and L_i	$\rho_{\mu}^{que}(k)$	Queue density in μ of i at k
$n_{sh,\mu}$	Constant of sampling instants related to v_{sh}	v_{sh}	Shockwave speed
$N_{i,\Sigma}^{in}(k)$	Total cumulative inflow of i at k	$v_{f,\mu}$	Free flow speed in μ of i
$N_{i,\Sigma}^{in,max}(k)$	Cumulative inflow limit of i at k	T_l	A shorter horizon of local layer
$N_{i,\Sigma}^{out}(k)$	Total cumulative link outflow of i at k	T_n	Prediction horizon of network layer
$N_{i,\Sigma}^{que}(k)$	Total cumulative queue inflow of i at k	$t_{f,\mu}$	Free- flow travel time
$N_{i,\mu}^{in}(k)$	Cumulative link inflow in μ of i at k	$\gamma_{L,\mu}$	Fraction of sampling instants related to $v_{f,\mu}$ and L_i
$N_{i,sh,\mu}^{in,max}(k)$	Maximum inflow with the shockwave traveling the queue backwards in μ of i at k	$\gamma_{f,\mu}$	Fraction of sampling instants related to $v_{f,\mu}$ and $L_{f,\mu}$
$N_{i,\mu}^{out}(k)$	Cumulative link outflow in μ of i at k	$\gamma_{sh,\mu}$	Fraction of sampling instants related to v_{sh}

with queue transmission to formulate the flow dynamics in the network. Finally, the controller design for the network and local layers of the MPC-Q is discussed in detail. Table I summarizes the description of the notations used in this study.

A. Framework of the MPC-Q

The proposed framework follows a hierarchical model predictive control structure with a network layer residing on top of a local layer as shown in Fig. 1. The network layer predicts the flow propagation in the considered network over a prediction horizon T_n and decides on the optimal green time fraction $\boldsymbol{b} = \left[b_1^{seg}, \ldots, b_m^{seg}\right]$ for each of the m road segments. The control objective of the network layer is to maximize network throughput and minimize oscillations in the signal plan. The LTM in this layer is able to capture flow dynamics without distinguishing the traffic states of different turning directions in a road segment.

The optimal green time fraction is sent to the local layer as a reference point. The local layer predicts the queue transmission for different turning movements at a shorter horizon $T_l < T_n$ and tracks the reference optimal green time fraction by optimally allocating the green time to different turning movements of a single intersection. The objective of the local layer is to decrease turn-level queue lengths at local intersections while maintaining network throughput optimized by the network layer.

We assume the network layer to receive segment-level inflow and outflow estimates as well as demand prediction, and the local controller receives turn-level inflow, queue inflow and outflow from a state estimation and prediction module.

B. Traffic Flow Dynamics

The road network is represented by a directed graph, which contains a set of road segments $\Pi = \Pi^O \cup \Pi^L$ connected by a set of nodes, where Π^O is the set of fictitious road segments at the origins that keep the demand; Π^L is the set of general road segments with multiple turning directions and $\Pi^O \cap \Pi^L = \varnothing$. It is assumed that the through and right-turn directions share the same lane, but our model can be easily extended to other lane configurations due to its link-based formulation. A road segment i of length L_i, combined with the adjacent upstream and downstream intersections, is divided into LT and TRT directions in this study. We assumed that the vehicle turning rates on each road segment are known.

Authorized licensed use limited to: Nanyang Technological University Library. Downloaded on September 10,2026 at 05:12:33 UTC from IEEE Xplore. Restrictions apply.

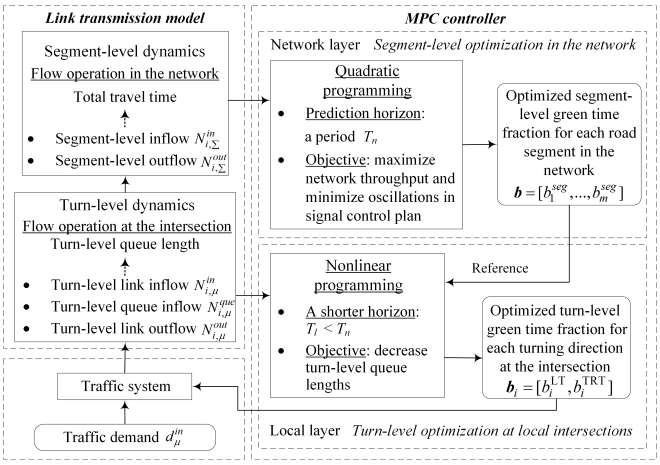


Fig. 1. Hierarchical framework of the proposed MPC-Q. $\boldsymbol{b}=\left[b_1^{seg},\ldots,b_m^{seg}\right]$ is the set of segment-level effective fractions of green time for all m road segments in the network; $\boldsymbol{b}_i=\left[b_i^{\mathrm{LT}},b_i^{\mathrm{TRT}}\right]$ is the set of turn-level effective fractions of green time for left-turn (LT) and through/right-turn (TRT) directions of road segment i, respectively. Note that the time index k is omitted from notations, but all the variables are time-dependent.

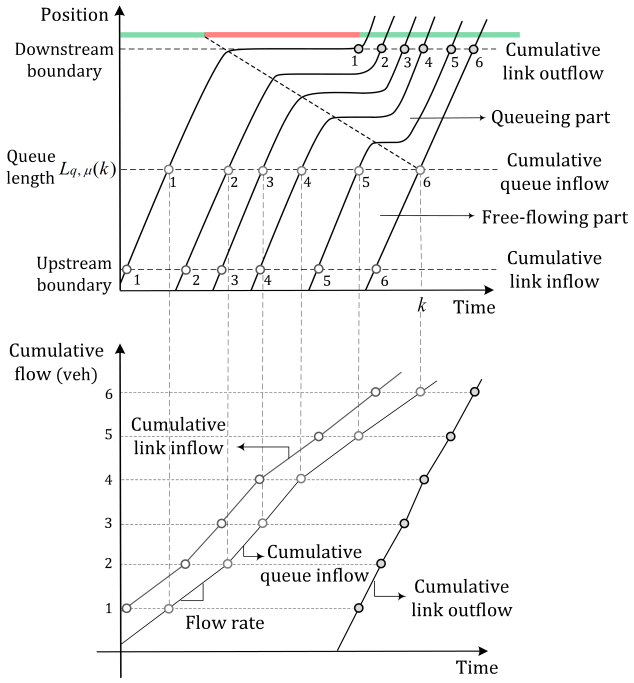


Fig. 2. Illustration of flow dynamics determined by the LTM with queue transmission. An road segment with length L_i is, for each in turning direction μ, divided into the free-flow and queueing parts of lengths $L_{f,\mu}(k)$ and $L_{q,\mu}(k)$, respectively.

We are now aiming at deriving state space representations for both, the local and the network layer. In the local layer, the flow dynamics are formulated in terms of cumulative turn-level queue inflow $N^{que}_{i,\mu}$, cumulative link inflow $N^{in}_{i,\mu}$ and outflow $N^{out}_{i,\mu}$ as shown in Fig. 2 which we will derive as follows:

1) Turn-Level Queue Inflow $N^{que}_{i,\mu}$: For the queue inflow estimation, the queue density $\rho^{que}_{\mu}(k)$ at time step k is firstly calculated based on the relationship between the flow rate and density, which can be characterized by a triangular fundamental diagram [27]:

$$\rho^{que}_{\mu}(k)=\rho^{c}_{\mu}+\frac{q^{c}_{\mu}-q^{out}_{i,\mu}(k)}{q^{c}_{\mu}}\left(\rho^{jam}_{\mu}-\rho^{c}_{\mu}\right), \tag{1}$$

where ρ^{jam}_{μ} represents the jam density of μ; ρ^{c}_{μ} and q^{c}_{μ} represent the critical density and flow rate; $q^{out}_{i,\mu}(k)$ represents the link outflow rate of μ at k. The lengths of the queueing part and the free-flow part of μ at k is given by:

$$L_{q,\mu}(k)=\left(N^{que}_{i,\mu}(k)-N^{out}_{i,\mu}(k)\right)/\rho^{que}_{\mu}(k), \tag{2}$$

$$L_{f,\mu}(k)=L_i-L_{q,\mu}(k), \tag{3}$$

Assuming that the free-flow speed $v_{f,\mu}$ is constant, the free-flow travel time can be calculated:

$$t_{f,\mu}=L_{f,\mu}(k)/v_{f,\mu}. \tag{4}$$

Since the decisions in signal control are implemented in a discrete manner, the flow dynamics need to be represented using discrete time intervals. Here, Δt represents the sampling time interval:

$$n_{f,\mu}=\lceil t_{f,\mu}/\Delta t\rceil, \tag{5}$$

$$\gamma_{f,\mu}=n_{f,\mu}-t_{f,\mu}/\Delta t. \tag{6}$$

where $n_{f,\mu}$ is the number of sampling time steps; γ_f represents the residual of sampling time steps. Since the mathematical operator $\lceil\cdot\rceil$ rounds the argument of the function to the nearest integer that is greater than the argument of the function, γ_f is thus in the range of [0,1]. Hence, the cumulative queue inflow of i is estimated by:

$$\begin{aligned}\boldsymbol{N}^{que}_{i}(k)=&(1-\boldsymbol{\gamma}_f)\boldsymbol{N}^{in}_{i}(k-\boldsymbol{n}_f)\\&+\boldsymbol{\gamma}_f\boldsymbol{N}^{in}_{i}(k+1-\boldsymbol{n}_f),\end{aligned} \tag{7}$$

where $\boldsymbol{N}^{que}_{i}(k)=[N^{que}_{i,\mathrm{LT}}(k),N^{que}_{i,\mathrm{TRT}}(k)]^{\top}$; $\boldsymbol{N}^{in}_{i}(k)=[N^{in}_{i,\mathrm{LT}}(k),N^{in}_{i,\mathrm{TRT}}(k)]^{\top}$; $\boldsymbol{\gamma}_f=[\gamma_{f,\mathrm{LT}},\gamma_{f,\mathrm{TRT}}]^{\top}$; $\boldsymbol{n}_f=[n_{f,\mathrm{LT}},n_{f,\mathrm{TRT}}]^{\top}$. Note that to simplify the formulation, we use $a-[b,c]^{\top}$ instead of $[a-b,a-c]^{\top}$; and $[a,b]^{\top}[c,d]^{\top}$ instead of $[ac,bd]^{\top}$.

If $\boldsymbol{\gamma}_f=[0,0]^{\top}$, the formulation of Eq. (7) can be simplified as the following:

$$\boldsymbol{N}^{que}_{i}(k)=\boldsymbol{N}^{in}_{i}\left(k-\boldsymbol{n}_f\right) \tag{8}$$

Consequently, the cumulative queue inflow at sampling time step k equals the cumulative link inflow at the n_f-th step before k as shown in Eq. (8). This indicates that the link inflow will be transferred to queue inflow in the future.

2) Turn-Level Link Inflow $N^{in}_{i,\mu}$: Considering a wave that propagates the outflow change upstream along the link, the maximum link inflow $N^{in,max}_{i,sh,\mu}$ with the shockwave traveling can be estimated [25]:

$$\begin{aligned}\boldsymbol{N}^{in,max}_{i,sh}(k)=&\boldsymbol{\gamma}_{sh}\boldsymbol{N}^{out}_{i}(k+1-\boldsymbol{n}_{sh})\\&+(1-\boldsymbol{\gamma}_{sh})\boldsymbol{N}^{out}_{i}(k-\boldsymbol{n}_{sh}),\end{aligned} \tag{9}$$

where $\boldsymbol{N}^{in,max}_{i,sh}(k)=[N^{in,max}_{i,sh,\mathrm{LT}}(k),N^{in,max}_{i,sh,\mathrm{TRT}}(k)]^{\top}$; $\boldsymbol{\gamma}_{sh}=[\gamma_{sh,\mathrm{LT}},\gamma_{sh,\mathrm{TRT}}]^{\top}$; $\boldsymbol{N}^{out}_{i}(k)=[N^{out}_{i,\mathrm{LT}}(k),N^{out}_{i,\mathrm{TRT}}(k)]^{\top}$; $\boldsymbol{n}_{sh}=[n_{sh,\mathrm{LT}},n_{sh,\mathrm{TRT}}]^{\top}$; $n_{sh,\mu}=\lceil(L_{q,\mu}(k)/v_{sh})/\Delta t\rceil$

Authorized licensed use limited to: Nanyang Technological University Library. Downloaded on September 10,2026 at 05:12:33 UTC from IEEE Xplore. Restrictions apply.

is the constant number of sampling instants; $\gamma_{sh,\mu} = n_{sh,\mu} - (L_{q,\mu}(k)/v_{sh})/\Delta t$ is the fraction of sampling instants.

The turn-leve link inflow limit $N_{i,\mu}^{in,max}(k+1)$ at $k+1$ includes the outflow-related released capacity and the maximum number of stored vehicles on the link, which can be estimated as follows:

$$N_{i,\mu}^{in,max}(k+1) = N_{i,sh,\mu}^{in,max}(k+1) + \rho_{q,\mu}(k)L_{q,\mu}(k) + \rho_{\mu}^{jam}L_{f,\mu}(k). \quad (10)$$

Therefore, the turn-level link inflow of road segment i from the links and origins directly upstream of i can be updated by Eq. (11), shown at the bottom of the next page, and should satisfy the constraint as shown in Eq. (12), at the bottom of the next page, where $q_{j,i}^{out}(k)$ is link outflow rate from road segment j to i that will be formulated in the following; $e_{i,\mu}$ is the turning rate; $d_{i,\mu}^{in}(k)$ is the input demand; $\boldsymbol{i}^{up}$ is the set of road segments directly upstream road segment i, respectively.

3) Turn-Level Link Outflow $N_{i,\mu}^{out}$: The turn-level maximum potential outflow $N_{i,\mu}^{out,max}(k)$ depends on the link inflow and can be estimated as follows:

$$\boldsymbol{N}_i^{out,max}(k) = \boldsymbol{\gamma}_L \boldsymbol{N}_i^{in}(k+1-\boldsymbol{n}_L) + (1-\boldsymbol{\gamma}_L)\boldsymbol{N}_i^{in}(k-\boldsymbol{n}_L), \quad (15)$$

where $\boldsymbol{N}_i^{out,max}(k) = [N_{i,\mathrm{LT}}^{out,max}(k), N_{i,\mathrm{TRT}}^{out,max}(k)]^{\top}$; $\boldsymbol{\gamma}_L = [\gamma_{L,\mathrm{LT}}, \gamma_{L,\mathrm{TRT}}]^{\top}$; $\boldsymbol{n}_L = [n_{L,\mathrm{LT}}, n_{L,\mathrm{TRT}}]^{\top}$; $n_{L,\mu} = \lceil (L_i/v_{f,\mu})\Delta t \rceil$ is the constant number of sampling instants; $\gamma_{L,\mu} = n_{L,\mu}(k) - (L_i/v_{f,\mu})\Delta t$ is the fraction of sampling instants.

If the downstream of road segment i has available supply, the turn-level desired cumulative outflow $N_{i,\mu}^{out,des}(k)$ of i at time step k limited by the green time and its link inflow that quantified by $N_{i,\mu}^{out,max}(k)$. Thus, the turn-level desired cumulative outflow $N_{i,\mu}^{out,des}(k)$ is estimated by:

$$\boldsymbol{N}_i^{out,des}(k) = \min\left(\boldsymbol{N}_i^{out}(k-1) + \boldsymbol{q}_i^s \boldsymbol{b}_i(k)\,\Delta t, \boldsymbol{N}_i^{out,max}(k)\right), \quad (16)$$

where $\boldsymbol{N}_i^{out,des}(k) = [N_{i,\mathrm{LT}}^{out,des}(k), N_{i,\mathrm{TRT}}^{out,des}(k)]^{\top}$; $\boldsymbol{q}_i^s = [q_{i,\mathrm{LT}}^s, q_{i,\mathrm{TRT}}^s]^{\top}$ is the saturation flow rate; $\boldsymbol{b}_i(k) = [b_i^{\mathrm{LT}}(k), b_i^{\mathrm{TRT}}(k)]^{\top}$; $b_i^{\mu}(k)$ is the turn-level effective fraction of green time of i at k, and $b_i^{\mu}(k) \in [0,1]$; The link outflow of i can thus be updated by Eq. (13), shown at the bottom of the next page, and should satisfy the constraint as shown in Eq. (14), at the bottom of the next page, where q_{cap} is the capacity of the origin.

After having derived the local layer dynamics, we are now in the position to conclude quickly the network layer dynamics as follows:

$$N_{i,\sum}^{in}(k) = \sum_{\mu} N_{i,\mu}^{in}(k) = \sum_{\mu} N_{i,\mu}^{in}(k-1) + \sum_{\mu} q_{i,\mu}^{in}(k)\Delta t, \quad (17)$$

where $N_{i,\sum}^{in}(k)$ is the total inflow of road segment i at time step k that can be calculated based on the turn-level link inflows; $q_{i,\mu}^{in}(k) = (N_{i,\mu}^{in}(k) - N_{i,\mu}^{in}(k-1))/\Delta t$ is the link inflow rate at k. Similarly, the total outflow of road segment i at time step k can be calculated as follows:

$$N_{i,\sum}^{out}(k) = \sum_{\mu} N_{i,\mu}^{out}(k) = \sum_{\mu} N_{i,\mu}^{out}(k-1) + \sum_{\mu} q_{i,\mu}^{out}(k)\Delta t, \quad (18)$$

where $q_{i,\mu}^{out}(k) = (N_{i,\mu}^{out}(k) - N_{i,\mu}^{out}(k-1))/\Delta t$ is the link outflow rate at k.

After having derived the dynamics in the local layer (Eqs. (11)-(14)) and the network layer (Eqs. (17)-(18)), respectively, we are now able to employ them in the corresponding MPC to predict the network flow propagation and turn-level queue length changes.

C. MPC-Q Controller Design

The extended LTM with queue transmission described in the previous section consists of state equations and constraints. This section details how these state equations and constraints can be included in the MPC so that the network throughput can be optimized while decreasing the queue lengths.

1) Network Layer: The primary objective of the network-layer controller is to maximize the network throughput while adhering to the flow dynamic equations outlined in Section II-B. Maximizing the network throughput can be realized by minimizing the total travel time spent [19], [25]. Existing methods usually take the independent green time at each simulation step as a control decision to optimize the travel time [19], [20], [25], which fails to guarantee the stability of the control plan, resulting in possible large oscillations in green time even over a short time interval. To address such limitation, this study minimizes the total travel time while considering the stability of the control plan, whereby formulating a quadratic programming (QP) problem with a quadratic objective function by involving the variations in decision variables. The objective still can be easily solved by any optimization solver, i.e., MATLAB, CPLEX.

Let $\Delta \boldsymbol{b}(k)$ be the variation in decision variables between successive time steps as follows:

$$\Delta \boldsymbol{b}(k) = \boldsymbol{b}(k) - \boldsymbol{b}(k-1), \quad (19)$$

and $\boldsymbol{b}(k) = \left[b_1^{seg}(k), \ldots, b_m^{seg}(k)\right]^{\top}$ be the set of system control inputs including segment-level green time fractions for all road segments in the network.

The system state is defined as the travel time spent, and represented as follows:

$$\boldsymbol{s}(k) = [s_1(k), s_2(k), \ldots, s_m(k)]^{\top}. \quad (26)$$

where $s_i(k) = \sum_{\mu} \Delta t (N_{i,\mu}^{in}(k) - N_{i,\mu}^{out}(k))$.

Then, the system state transition function is given by:

$$\boldsymbol{s}(k+1) = \boldsymbol{s}(k) + \boldsymbol{B}(k)\boldsymbol{b}(k) + \boldsymbol{C}(k)\boldsymbol{d}^{in}(k). \quad (27)$$

Eq. (27) indicates that the system state transition at $k+1$ relates to the current system state $\boldsymbol{s}(k)$ and the external disturbances at k caused by the input demands $\boldsymbol{d}^{in}(k) = [d_{1,\mathrm{LT}}^{in}(k) + d_{1,\mathrm{TRT}}^{in}(k), \ldots, d_{m,\mathrm{LT}}^{in}(k) + d_{m,\mathrm{TRT}}^{in}(k)]^{\top}$ and the signal control

Authorized licensed use limited to: Nanyang Technological University Library. Downloaded on September 10,2026 at 05:12:33 UTC from IEEE Xplore. Restrictions apply.

$\boldsymbol{b}(k)$. $\boldsymbol{B}(k) \in \mathbb{R}^{m\times m}$ and $\boldsymbol{C}(k) \in \mathbb{R}^{m\times m}$ can be easily derived by the technical transformation of Eqs. (11), (13) and (26), and thus include saturation flow rates, the discrete-time step, and turning rates.

Let $\boldsymbol{S}_k \in \mathbb{R}^{\omega\times 1}$ and $\boldsymbol{U}_k \in \mathbb{R}^{\omega\times 1}$ be the state and control vectors during the prediction horizon, respectively, where $\omega = (T_n+1)m$ is the number of effective fractions of green time at the segment level over the prediction horizon. We can have:

$$\boldsymbol{S}_k = [\boldsymbol{s}(k|k), \boldsymbol{s}(k+1|k), \ldots, \boldsymbol{s}(k+T_n|k)]^\top, \tag{28}$$

$$\boldsymbol{U}_k = [\boldsymbol{b}(k|k), \boldsymbol{b}(k+1|k), \ldots, \boldsymbol{b}(k+T_n|k)]^\top, \tag{29}$$

$\boldsymbol{S}_k$ can also involve the difference between the predicted and actual states to ensure control performance in the presence of model errors. The idea now is to find an optimal control vector $\boldsymbol{U}_k$ to minimize the sum of $\boldsymbol{S}_k \in \mathbb{R}^{\omega\times 1}$ over the prediction horizon under the given initial system state $\boldsymbol{s}(k)$ considering the constraints on flow dynamics and signal timing. We can thus formulate the QP-based optimization problem in the network layer as shown in Eqs. (20)-(25), at the bottom of the page, where $i \in \Pi$, $j \in \Pi^O$ and $\forall k \in \{1, \ldots, T_n\}$, and the state vector $\boldsymbol{S}_k$ is determined by Eqs. (26)-(28); $\boldsymbol{I}^c$ represents the set of signal phases whose moving directions conflict with each other, thus the constraint of Eq. (24) ensures that the fraction assignment of green time does not have overlapping parts in conflict phases. The variations in green time fractions and travel times have different scales. Therefore, this study assigns weights $\boldsymbol{Q}$ and $\boldsymbol{R}$ for these two quantities. We just set $\boldsymbol{Q} = \boldsymbol{R} = diag\{1, 1, \ldots 1\}$. Thus, the second term of the objective, which involves the minimization of the variations in green fractions, only takes effect when the first cost term of travel time $\boldsymbol{S}_k$ is reduced to the same scale. It should be clarified that a quadratic form increases the penalty for larger deviations in travel time. For instance, small increases in travel time have a smaller impact on the objective function, while large increases have a disproportionately larger effect, making it easier to find optimal solutions under congested conditions.

Therefore, our approach transform the conventional node model of LTMs into some optimization constraints as defined in Eqs. (21)-(23), which allows connecting the road segments of the road network in a simple but effective way and facilitates the resolution of MPC-Q. It's worth noting that although some MPC methods using LTMs without the node model have been proposed [19], [25], these methods still may violate the constraints in Eqs. (21)-(23) due to their inadequate consideration of turn-level queue dynamics within the roads.

2) Local Layer: The outputs of the network-layer controller are the optimized segment-level effective fractions of green time, as demonstrated in Eq. (20). The control task of the local layer is to assign the desired segment-level effective fraction of green time to each turning direction of the corresponding intersection based on the predicted turn-level queue lengths over the horizon T_l. The constraints are determined depending on whether the intersection has protected left-turn phases as shown in Fig. 3.

a) Protected left-turn phases exist: Let δ and $\bar{\delta}$ denote the incoming road segments from the southern and northern directions or western and eastern directions of the intersection, respectively. We make the assumption that the green time for

$$N_{i,\mu}^{in}(k+1) = \begin{cases} N_{i,\mu}^{in}(k) + \sum_{j\in \boldsymbol{i}^{up}} q_{j,i}^{out}(k) e_{i,\mu}\Delta t & i \in \Pi^L \\ N_{i,\mu}^{in}(k) + d_{i,\mu}^{in}(k)\Delta t & i \in \Pi^O \end{cases} \tag{11}$$

$$N_{i,\mu}^{in}(k+1) \le N_{i,sh,\mu}^{in,max}(k+1) + \rho_{q,\mu}(k) L_{q,\mu}(k) + \rho_{\mu}^{jam} L_{f,\mu}(k) \tag{12}$$

$$N_{i,\mu}^{out}(k+1) = \begin{cases} N_{i,\mu}^{out}(k) + b_i^{\mu}(k) q_{i,\mu}^{s}\Delta t & i \in \Pi^L \\ N_{i,\mu}^{out}(k) + b_i^{\mu}(k) q_{cap}\Delta t & i \in \Pi^O \end{cases} \tag{13}$$

$$N_{i,\mu}^{out}(k+1) \le \begin{cases} \min(N_{i,\mu}^{out}(k) + q_{i,\mu}^{s} b_i^{\mu}(k)\Delta t, N_{i,\mu}^{out,des}(k+1)) & i \in \Pi^L \\ \min(N_{i,\mu}^{out}(k) + q_{cap} b_i^{\mu}(k)\Delta t, N_{i,\mu}^{out,des}(k+1)) & i \in \Pi^O \end{cases} \tag{14}$$

$$\min_{\boldsymbol{U}_k} \quad \boldsymbol{S}_k^\top \boldsymbol{Q} \boldsymbol{S}_k + \Delta \boldsymbol{U}_k^\top \boldsymbol{R} \Delta \boldsymbol{U}_k \tag{20}$$

$$s.t. \quad N_{i,\Sigma}^{in}(k+1) - \left(N_{i,sh,\Sigma}^{in,max}(k+1) + \sum_{\mu} \left(\rho_{q,\mu}(k) L_{q,\mu}(k) + \rho_{jam,\mu}(k) L_{f,\mu}(k) \right) \right) \le 0, \tag{21}$$

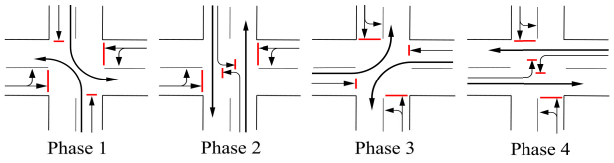
$$N_{i,\Sigma}^{out}(k+1) - \min\left(N_{i,\Sigma}^{out}(k) + b_i^{seg}(k) \sum_{\mu} q_{i,\mu}^{s}\Delta t, N_{i,\Sigma}^{out,max}(k+1) \right) \le 0, \tag{22}$$

$$N_{j,\Sigma}^{out}(k) - N_{j,\Sigma}^{in}(k) \le 0, \tag{23}$$

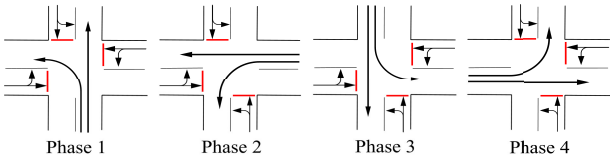
$$\sum_{b_i^{seg}\in \boldsymbol{I}^c} b_i^{seg}(k) \le 1, \tag{24}$$

$$0 \le b_i^{seg}(k) \le 1. \tag{25}$$

Authorized licensed use limited to: Nanyang Technological University Library. Downloaded on September 10,2026 at 05:12:33 UTC from IEEE Xplore. Restrictions apply.



(a) Protected left-turn phase exist.



(b) Protected left-turn phase do not exist.

Fig. 3. Illustration of signal phases at the local intersection.

the LT or TRT in the southern direction is equal to that in the northern direction, and the same holds for the western and eastern directions.

Let $\boldsymbol{b}_\mu = \left[\boldsymbol{b}_\mu(l), \boldsymbol{b}_\mu(l+1), \ldots, \boldsymbol{b}_\mu(l+T_l)\right]^\top$ be the set of turn-level effective fractions of green time of all incoming road segments over the horizon T_l. The nonlinear programming (NLP) problem at the local layer can be expressed as:

$$\min_{\boldsymbol{b}_\mu} \sum_{\forall l \in T_l} \sum_i \sum_\mu \left| b_i^\mu(l) - b_i^{seg}(l) \frac{L_{q,\mu}^i(k)}{\sum_\mu L_{q,\mu}^i(k)} \right| \tag{30}$$

$$s.t. \quad \text{Eq. (12), Eq. (14)},$$

$$b_\delta^{\mathrm{LT}}(l) - b_{\bar{\delta}}^{\mathrm{LT}}(l) = 0, \tag{31}$$

$$b_\delta^{\mathrm{TRT}}(l) - b_{\bar{\delta}}^{\mathrm{TRT}}(l) = 0, \tag{32}$$

$$0 \le b_i^\mu(l) \le 1. \tag{33}$$

The formulation of NLP may become computationally intensive. Therefore, to reduce the computational overhead, instead of setting the initial values of the decision variables in the local layer (e.g., turn-level green time fractions) to numerous random values, the NLP optimization is warm-started using predefined initial values as shown in Eq. (34). These initial values are determined by assigning the segment-level green time fractions to the corresponding turning directions based on the queue lengths. Moreover, the local layer operates with distributed control at each intersection rather than centralized control. Its objective is to manage the queue length at individual intersections within a shorter horizon (e.g., 60s in this study), eliminating the need to measure flow dynamics across the entire network and thereby reducing the computational load of the NLP. Consequently, the complexity of the local-layer NLP remains unchanged as the number of controlled intersections increases. Note that the real-time feasibility in this study refers to the MPC calculation time being shorter than the model's sampling time interval. The proposed MPC utilizes the macroscopic link-based flow model, which allows for a large sampling time interval to predict flow dynamics of 10s and hence a maximum of 6 prediction steps.

$$[b_i^{\mu,0}(1), \ldots, b_i^{\mu,0}(T_l)] = [b_i^{seg}(1) \frac{L_{q,\mu}^i(k)}{\sum_\mu L_{q,\mu}^i(k)}, \ldots, b_i^{seg}(T_l) \frac{L_{q,\mu}^i(k)}{\sum_\mu L_{q,\mu}^i(k)}] \tag{34}$$

b) Protected left-turn phases do not exist: If there are no protected left-turn phases, we do not need to constrain the green times for related turning directions, as shown in Eqs. (31) and (32). The optimized effective fractions of green time at each intersection can be obtained by solving Eq. (30) constrained by Eq. (12), Eq. (14), and Eq. (33). The problems formulated in Eq. (20) and Eq. (30) can be easily solved by any optimization solver.

III. Simulation Experiments

This section presents numerical simulation results showing the performance of the proposed method for network control under different traffic conditions. Two case studies are considered: (1) An arterial with two successive intersections to demonstrate in detail the ability of the proposed MPC-Q to control the queues in the presence of spillbacks; (2) A road network is used to demonstrate the performance of the proposed MPC-Q under different traffic regimes and to compare it with state-of-the-art methods.

A. Arterial With Successive Intersections

The simulated arterial represents a typical urban signalized arterial structure, consisting of 14 road segments, 6 origins, and 6 destinations, as shown in Fig. 4. Each node's incoming road segment is divided into LT and TRT directions. Table II describes the network parameters for the simulation. The road lengths can be easily adjusted for different implementation locations since the proposed MPC is link-based and does not require spatial discretization within the roads. In Table II, the decreased maximum outflow of D_4 is set to trigger congestion within the network. The traffic flow parameters are set to reasonable default values.

The time-varying traffic demand inputs for each turning direction of the origins are presented in Fig. 5. Taking the TRT direction as an example, the first 250 s serve as the warm-up stage, after which the input demand is increased to simulate the traffic peak period until the 620 s. After the 620 s, the demand gradually decreases. The turning rates for other general links are presented in Fig. 4.

To demonstrate the superiority of MPC-Q considering the impacts of queue transmission on the network control, this section chooses the MPC without that as a benchmark method (MPC-B) [25] to compare with our MPC-Q. We use a macroscopic cell transmission model in combination with a supply-proportional node model as the system model to evaluate the performance of MPC-Q and MPC-B. This system model is different from the extended or standard LTM but ensures computational efficiency. Since the sampling interval is set to 10 s and free-flow speed is 10 m/s, the cell length is 100 m. The horizons of the network-layer T_n and the local-layer T_l are set to 600 s and 60 s, respectively. The impacts of T_n and T_l settings will be discussed later.

We will evaluate the proposed method by using density and queue length. Fig. 6 shows the evolution of space-time average density at three bottleneck-realted paths associated with MPC-Q and MPC-B. The intensity of the color reflects the level of congestion, with redder shades indicating higher congestion

Authorized licensed use limited to: Nanyang Technological University Library. Downloaded on September 10,2026 at 05:12:33 UTC from IEEE Xplore. Restrictions apply.

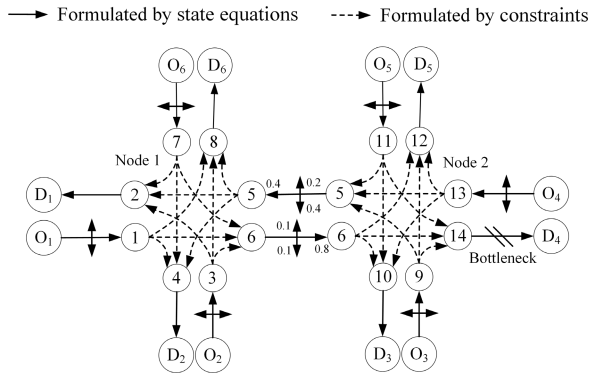


Fig. 4. Illustration of the arterial with successive intersections, where O_1 to O_6 represent the origins; D_1 to D_6 represent the destinations; the bottleneck is located on road segment 14; the paths related to the bottleneck are 1-6-14, 9-14, and 11-14.

TABLE II
ROAD SEGMENT PARAMETERS FOR THE SIMULATED SIGNALIZED ARTERIAL

Length (m)						
L_1	L_2	L_3	L_4	L_5	L_6	L_7
600	600	300	300	200	200	300
L_8	L_9	L_{10}	L_{11}	L_{12}	L_{13}	L_{14}
300	400	400	300	300	300	300
Maximum outflow of the destination (veh/h)						
D_1	D_2	D_3	D_4	D_5	D_6	
2000	2000	2000	200	2000	2000	
Traffic flow parameters (applied for all road segments)						
Saturation flow rate	2000 (veh/h)					
Shockwave speed	15 (m/s)					
Jam density	160 (veh/km)					
Free-flow speed	10 (m/s)					

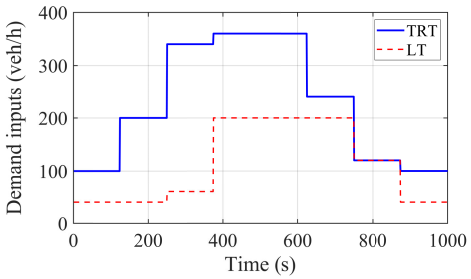
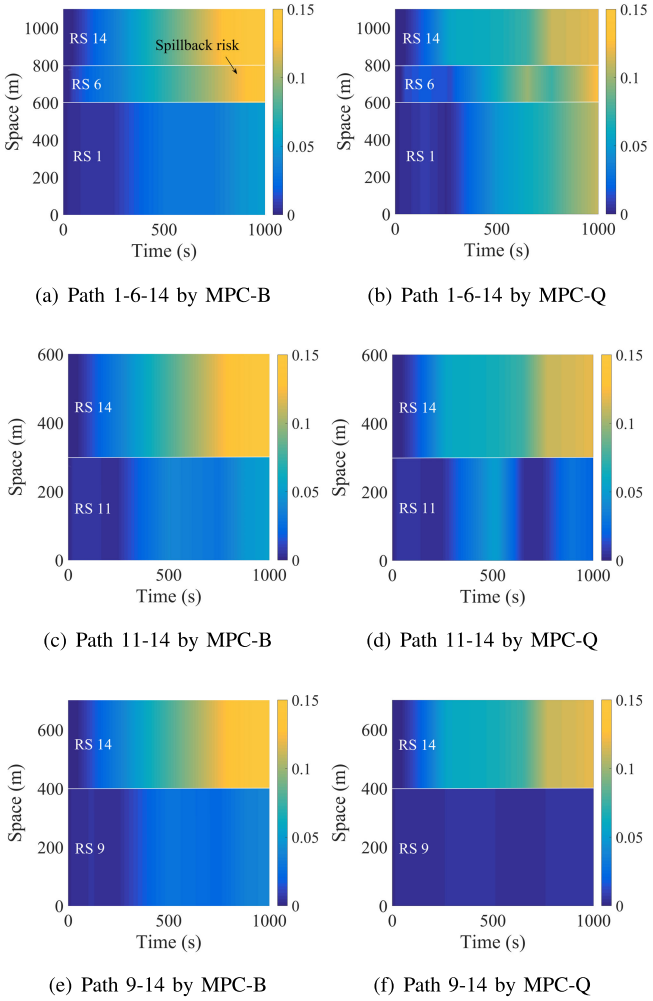


Fig. 5. Demand inputs for each turning direction at the origins O_1 to O_6.



(a) Path 1-6-14 by MPC-B (b) Path 1-6-14 by MPC-Q

(c) Path 11-14 by MPC-B (d) Path 11-14 by MPC-Q

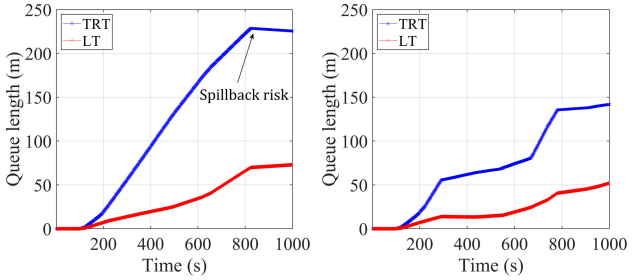
(e) Path 9-14 by MPC-B (f) Path 9-14 by MPC-Q

Fig. 6. Evolution of space-time average density (veh/m) on bottleneck-related paths using MPC-B and MPC-Q.

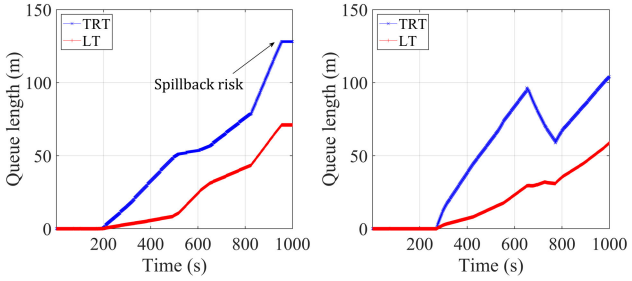
levels. It is clear that the proposed MPC-Q eliminates the spillbacks in Path 1-6-14, 9-14 and 11-14 caused by the outflow drop at the bottleneck on road segment 14. The baseline MPC-B failed to eliminate queue spillback along the path RS 14-6-1 because it uses the standard LTM to predict flow dynamics. With the spillback dissipating on road segment 14, there is a slight increase in density from road segments 1 to 6, but this can be acceptable because the density distribution along Path 1-6-14 eventually becomes uniform as shown in Fig. 6b. Fig. 7 compares the detailed turn-level queue length change at the road segments along Path 1-6-14 before and after optimization. Since the MPC-B only focuses on optimizing network throughput, the queue length cannot be controlled and obtained by this approach, thus only the proposed MPC-Q is compared against uncontrolled results in Fig. 7. The queue length of the link with spillback risk can be reduced after the control optimization.

Fig. 8 presents the control stability in terms of the effective fraction of green time associated with MPC-B and MPC-Q during the peak period. Since the findings are similar for different nodes' incoming road segments, only the results of road segment 1 are presented for analysis. To ensure a fair comparison, both MPC-Q and MPC-B are assigned a left-turn protected phase. Note that, since the flow model in MPC-B operates at the segment level without distinguishing between flow operations in different turning directions, we multiply the segment-level aggregated green time fraction by the turning rates to obtain the turn-level fractions when using MPC-B. As shown in Figs. 6 and 8, the proposed MPC-Q provides effective green time fractions for different turning directions, to eliminate spillback risks. MPC-Q finds more stable control solutions that avoid the large oscillations in green time observed in MPC-B (e.g., 400 s to 550 s in Fig. 8), which obviously benefits road safety and fuel consumption. The reason for this is that the MPC-Q utilizes quadratic programming with the variations of controller decision variables to

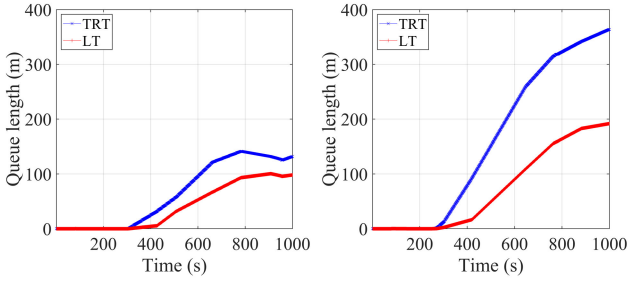
Authorized licensed use limited to: Nanyang Technological University Library. Downloaded on September 10,2026 at 05:12:33 UTC from IEEE Xplore. Restrictions apply.



(a) Before optimization: RS 14 (b) After optimization: RS 14



(c) Before optimization: RS 6 (d) After optimization: RS 6



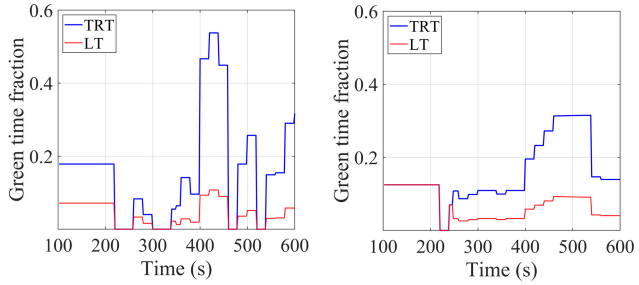
(e) Before optimization: RS 1 (f) After optimization: RS 1

Fig. 7. Comparison of turn-level queue length changes at Path 1-6-14 before and after the optimization, were RS represents road segment.

include the control stability, while MPC-B chooses the linear programming with the controller decision variable at each time step, thus failing to quantify the impacts of the variations in signal timing on the network operation over the prediction horizon.

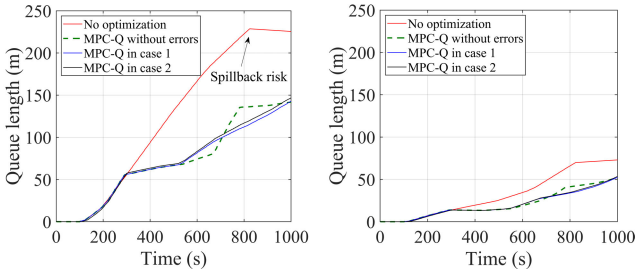
Considering that the varying vehicle sizes and driver behaviors impact the free-flow speed, we set different free-flow speeds for the system and prediction models in the proposed MPC framework. This adjustment introduces discrepancies in the prediction model compared to real traffic. The model sensitivity can thus be analyzed. The actual free-flow speed is assumed to be 10 m/s. Two cases on the arterial with consecutive intersections as shown in Fig. 4 are performed for comparison: a) case 1: the free-flow speed for the prediction model is set to 12 m/s, larger than the actual value; b) case 2: the free-flow speed for the prediction model is set to 8 m/s, smaller than the actual value. The comparison results are shown in Figs. 9 and 10.

The proposed MPC framework can still mitigate spillback risks. The queue lengths at the bottleneck link as shown in Fig. 9 remain similar across different cases. The proposed model can address certain noise issues for the following



(a) MPC-B (b) MPC-Q

Fig. 8. Comparison of control stability in terms of effective fraction of green time for road segment 1 associated with MPC-B and MPC-Q.



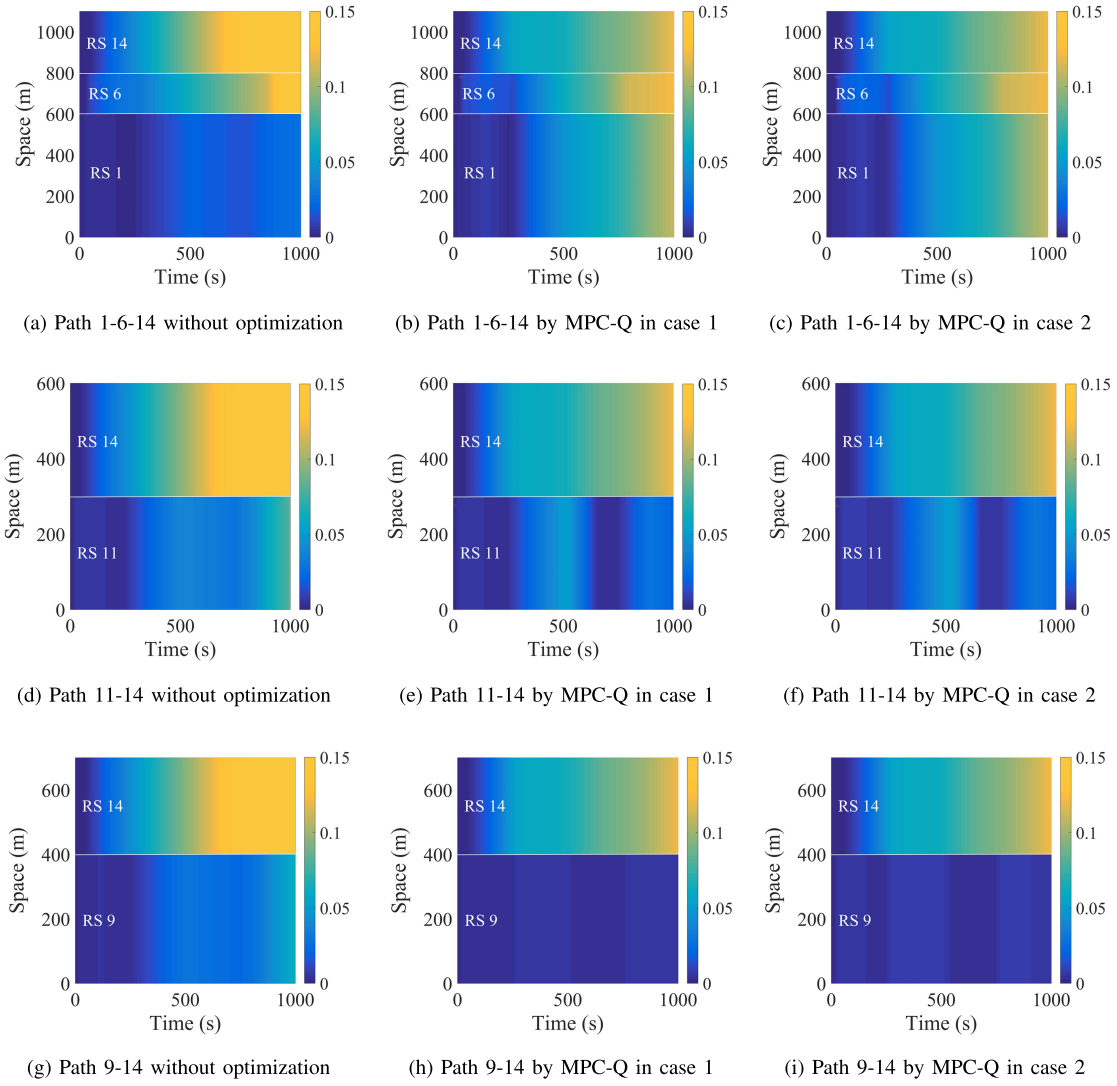
(a) (a) TRT direction (b) (b) LT direction

Fig. 9. Comparison results of queue length at bottleneck link 14 under model errors.

reasons: Firstly, the proposed LTM-based MPC does not require sophisticated spatiotemporal discretization, allowing for a larger sampling time interval to predict flow dynamics, which helps smooth fluctuations caused by noise, such as Gaussian white noise in the demand data. Secondly, by updating control actions based on time-aggregated flow dynamics, the MPC can reduce the impact of time-lagged flow changes between predicted and actual traffic conditions caused by noise.

Then, Table III summarizes the computational performance of the proposed MPC-Q, where network throughput is defined as $\mathcal{O} = \frac{3600}{T}\sum_i^m \sum_\mu N_{i,\mu}^{out}(T)$. Simulations were carried out by MATLAB on a laptop with an Intel i7-8550U CPU 1.80 GHz and 16GB RAM. From Table III, we can find that increasing the prediction horizon results in increased computational time, but may without further improved network throughput. Therefore, a longer local-layer prediction horizon does not always correlate with better control performance. The prediction horizon should be determined based on the specific requirements of each case. Note that the network layer predicts the flow propagation over T_n and decides on the optimal green time fraction for each road segment. The optimal green time fraction is sent to the local layer as a reference point. To provide an effective reference, T_l should be smaller than T_n. Consequently, only the computational time for the case where $T_l < T_n$ is compared in Table III. In this case, the greatest computational time is still smaller than the sampling interval of 10 s, thus the proposed MPC-Q can be real-time feasible.

Authorized licensed use limited to: Nanyang Technological University Library. Downloaded on September 10,2026 at 05:12:33 UTC from IEEE Xplore. Restrictions apply.



(a) Path 1-6-14 without optimization (b) Path 1-6-14 by MPC-Q in case 1 (c) Path 1-6-14 by MPC-Q in case 2

(d) Path 11-14 without optimization (e) Path 11-14 by MPC-Q in case 1 (f) Path 11-14 by MPC-Q in case 2

(g) Path 9-14 without optimization (h) Path 9-14 by MPC-Q in case 1 (i) Path 9-14 by MPC-Q in case 2

Fig. 10. Evolution of space-time average density (veh/m) on bottleneck-related paths by MPC-Q with model errors.

TABLE III

OVERVIEW OF THE COMPUTATIONAL TIME $\mathcal{T}$ (CPUS) PER STEP AND NETWORK THROUGHPUT $\mathcal{O}$ (VEH/H) UNDER DIFFERENT MPC-Q PREDICTION HORIZONS

Network-layer T_n	Local-layer T_l											
	10s		20 s		30 s		40 s		50 s		60 s	
	$\mathcal{O}$	$\mathcal{T}$	$\mathcal{O}$	$\mathcal{T}$	$\mathcal{O}$	$\mathcal{T}$	$\mathcal{O}$	$\mathcal{T}$	$\mathcal{O}$	$\mathcal{T}$	$\mathcal{O}$	$\mathcal{T}$
10 s	852	0.08	-	-	-	-	-	-	-	-	-	-
20 s	854	0.08	858	0.08	-	-	-	-	-	-	-	-
30 s	920	0.08	953	0.08	953	0.09	-	-	-	-	-	-
40 s	1032	0.08	1052	0.08	1055	0.09	1055	0.09	-	-	-	-
50 s	1041	0.08	1050	0.08	1052	0.09	1058	0.09	1092	0.09	-	-
60 s	1140	0.08	1142	0.08	1149	0.09	1162	0.09	1162	0.09	1162	0.11
80 s	1140	0.08	1149	0.08	1151	0.09	1162	0.09	1163	0.09	1163	0.14
100 s	1142	0.08	1150	0.08	1153	0.09	1162	0.1	1163	0.1	1163	0.15
150 s	1142	0.1	1148	0.11	1153	0.12	1162	0.13	1163	0.15	1163	0.25
200 s	1142	0.12	1145	0.14	1153	0.15	1162	0.17	1163	0.22	1163	0.34
300 s	1142	0.17	1148	0.2	1153	0.28	1162	0.25	1163	0.37	1163	0.7
400 s	1142	0.32	1147	0.37	1153	0.42	1162	0.54	1163	0.77	1163	1.64
500 s	1142	0.44	1150	0.62	1153	0.74	1162	0.82	1163	1.18	1163	2.41
600 s	1142	0.68	1148	0.79	1153	0.99	1162	1.19	1163	1.82	1163	3.86

B. Road Network

This section compares the control performance of the proposed MPC-Q in an urban road grid network under different traffic regimes. To this end, we generate a simulated network consisting of 48 road segments as depicted in Fig. 11. Bottlenecks are assumed downstream of road segment 13 and 36,

Authorized licensed use limited to: Nanyang Technological University Library. Downloaded on September 10,2026 at 05:12:33 UTC from IEEE Xplore. Restrictions apply.

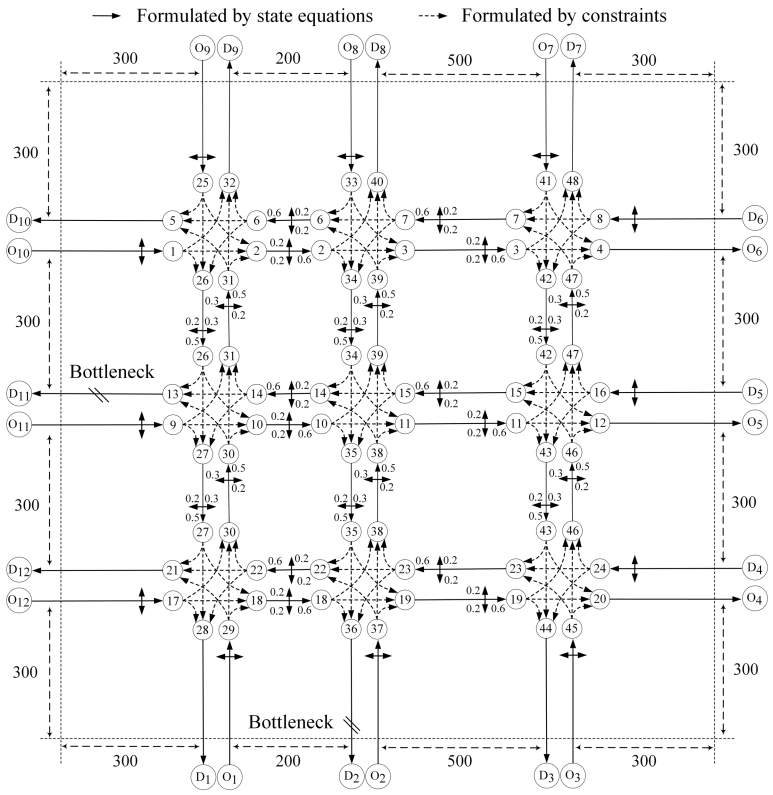


Fig. 11. Illustration of the simulated grid network.

TABLE IV
ROAD SEGMENT PARAMETERS FOR THE SIMULATED URBAN ROAD NETWORK

Maximum outflow of the destination (veh/h)					
D_1	D_2	D_3	D_4	D_5	D_6
2000	200	2000	2000	2000	2000
D_7	D_8	D_9	D_{10}	D_{11}	D_{12}
2000	2000	2000	2000	200	2000
Traffic parameters (applied for all road segments)					
Saturation flow rate	2000 (veh/h)				
Shockwave speed	15 (km/h)				
Jam density	160 (veh/km)				
Free-flow speed	10 (m/s)				

respectively. Table IV presents the network parameters. The simulation experiments are analyzed for 1000 s using the same network parameters with only differences in traffic regimes:

- Symmetrical undersaturated (traffic regime 1): 300 veh/h for TRT and 100 veh/h for LT in all directions including northern, southern, western and eastern directions.

- Asymmetric undersaturated (traffic regime 2): 400 veh/h for TRT and 200 veh/h for LT in the northern/southern direction; 150 veh/h for TRT and 50 veh/h for LT in the western/eastern direction.

- Symmetrical oversaturated (traffic regime 3): 1500 veh/h for TRT and 500 veh/h for LT in all directions.

- Asymmetric near-oversaturation (traffic regime 4): 250 veh/h for TRT and 150 veh/h for LT in the northern/southern direction; 1500 veh/h for TRT and 500 veh/h for LT in the western/eastern direction.

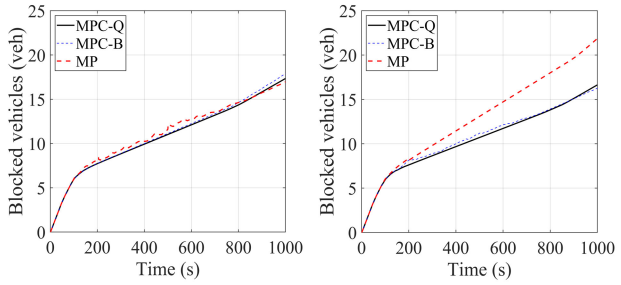
It should be clarified that, in this study, we assume the network has bottlenecks with reduced maximum allowable outflow. The conditions of being oversaturated or undersaturated are determined by whether there are spillback risks at the bottleneck link. We assumed a bottleneck exists in the network as shown in Fig. 4. The MPC-Q controller aims to prevent bottleneck-related spillback at link 14. If spillback occurs, it significantly will reduce network throughput, so the MPC-Q will hold inflow to the bottleneck and allow more green time to long-queue links. The controller works as long as there is buffer space in the network to store queues. However, in gridlock situations, the current design is likely ineffective and the objective function should be refined to deal with this. Note that the vehicle turning rates on the origin links, e.g., links 10, 11, 17, 29, 37, and 45 in Fig. 11, vary under different traffic regimes. These turning rates can be calculated based on the turn-level input demands. For example, in symmetrical undersaturated conditions (traffic regime 1), the vehicle turning rates for LT and TRT are 0.25 and 0.75, respectively.

We compare our MPC-Q with the following methods: (a) the MPC-B approach [25]; (b) a max-pressure-based approach (MP) [35] for finding the dynamic signal timings to maximize the network throughput. This approach is chosen because it considers the impacts of the queueing process on the intersection control, which is associated with the pressure for each phase. Since the MPC-B only considers the link inflow and outflow at road segment boundaries, the queue length can not be obtained. We define the blocked vehicles at time step k as $Aq = \frac{1}{m}\sum_{i=1}^{m}\sum_{\mu}\left(N_{i,\mu}^{in}(k) - N_{i,\mu}^{out}(k)\right)$, to perform the

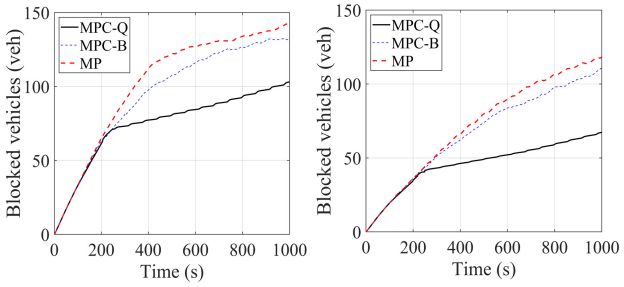
Authorized licensed use limited to: Nanyang Technological University Library. Downloaded on September 10,2026 at 05:12:33 UTC from IEEE Xplore. Restrictions apply.

TABLE V
OVERVIEW OF THE COMPARISON RESULTS BY DIFFERENT METHODS

Method	MPC-Q	MPC-B	MPC-Q vs. MPC-B (%)	MP	MPC-Q vs. MP (%)
Traffic regime 1					
Average blocked vehicles (veh)	10	10	0	10	0
Average throughput (veh/h)	382	380	0	380	0
Traffic regime 2					
Average blocked vehicles (veh)	9	10	-10	12	-25
Average throughput (veh/h)	386	385	0	380	0
Traffic regime 3					
Average blocked vehicles (veh)	62	88	-29	102	-39
Average throughput (veh/h)	540	498	8	426	26
Traffic regime 4					
Average blocked vehicles (veh)	35	65	-46	78	-55
Average throughput (veh/h)	521	480	9	432	21

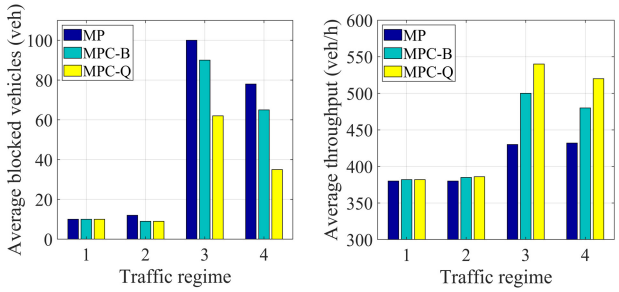


(a) Traffic regime 1 (b) Traffic regime 2



(c) Traffic regime 3 (d) Traffic regime 4

Fig. 12. Evolution of blocked vehicles Aq for varied demand profiles with different methods.



(a) Average blocked vehicles (b) Average throughput

Fig. 13. Comparison of average blocked vehicles and throughput under different traffic regimes using different control methods.

comparison. The higher the Aq is, the greater the congestion and the longer the travel time spent. The sampling intervals in MPC-B and MPC-Q are all set to 10 s. After comparing the results under different horizon values, we set T_n and T_l to 100 s and 60 s, respectively, to achieve the cost-efficient control purpose in this case.

Fig. 12 describes the evolution of the blocked vehicles Aq under four traffic regimes using different methods. We observe that the performance of the three compared control approaches does not have obvious differences at the first 100s. During the simulation process, the variations between them become more and more apparent due to the increased network loads, especially under traffic regimes 3 and 4. Specifically, Aq increases slowly under traffic regimes 1 and 2, and the performance of MPC-Q and MPC-B is similar and better than that of MP. In dense traffic conditions, i.e., regimes 3 and 4 the blocked vehicles Aq associated with MPC-B and MP, respectively is always higher than the one for MPC-Q. Thus, we conclude that MPC-Q provides more effective signal control plans that decreases the number of queueing vehicles in the oversaturated conditions.

Fig. 13 and Table V provide an overview of the comparison results of MPC-Q, MPC-B and MP, where the average blocked vehicles is the mean value of Aq over the simulation period T. While for traffic regimes 1 and 2 the differences between the average blocked vehicles for the different control strategies are neglectable, they are significant for traffic regimes 3 and 4 (cf. Fig. 13). In regimes 3 and 4, MPC-Q reduced the average blocked vehicles by at least 29%. The comparison of the throughput demonstrates similar improvements when utilizing MPC-Q (cf. Table V). Hence, incorporating the signal control impacts on the network's future operation, both MPC-B and MPC-Q achieve better performance compared to MP. However, the lack of consideration of the queue transmission in MPC-B is unfavorable to effectively representing the queue length change under saturated conditions, resulting in lower average throughput and larger blocked vehicles compared to MPC-Q. Thus, the proposed MPC-Q achieves the most desirable performance.

Simulations were carried out by MATLAB on a laptop with an Intel i7-8550U CPU 1.80 GHz and 16GB RAM. The average computational time for each simulation step of the proposed MPC-Q under different regimes is 0.35 CPU-s, which is obviously smaller than the LTM sampling interval. Hence, the MPC-Q is still real-time feasible in the simulated grid network with nine intersections.

Authorized licensed use limited to: Nanyang Technological University Library. Downloaded on September 10,2026 at 05:12:33 UTC from IEEE Xplore. Restrictions apply.

IV. Conclusion

This study proposed a novel MPC approach that uses the LTM with queue transmission to optimize the traffic signals in urban road networks. In the network layer, a QP-based controller with inputs of the decision variations was formulated to improve the network-wide throughput. By converting the generally used time-step independent signal timing into signal timing variations, the proposed MPC-Q framework is capable of considering signal control stability. In the local layer, an NLP-based controller with reference tracking was developed to track and refine the segment-level effective fractions of green time output by the network layer to the turn level. Thus, the proposed MPC-Q can be performed without assuming that flow operations in different turning directions are homogeneous. Both arterial and network simulation experiments were performed to demonstrate the performance of the proposed algorithms. The obtained simulation results have demonstrated the superiority of the proposed framework compared to the existing rival methods.

The main challenge is how to incorporate the traffic variations in turn movements within the MPC, while ensuring that the MPC has less computational overheads to be real-time feasible. To this end, the LTM with turn-level link inflow, queue inflow and link outflow was integrated into a hierarchical MPC framework, which enables the controller to capture the detailed flow propagation in multiple turning directions within the road segment without imposing complex computational overheads on the network-layer coordination, since the computational overheads can be reduced by aggregating turn-level flow dynamics into segment levels at the network layer. Moreover, in each road segment, the continuous turn-level effective fractions of green time are used as controller decision variables instead of widely used binary values, e.g., 0 for green and 1 for red, which eliminates the need for solving the cumbersome MILP problems and enables MPC to generate fine-grained yet real-time feasible outputs. Compared to the existing MPC-based methods using LTMs [19], [25], the proposed MPC-Q method addresses the limitation of being difficult to adapt to traffic variations at turn level in large networks and accounts for the queue transmission.

Urban traffic involves multiple vehicle types, e.g., buses, bicycles, and trams, but the proposed MPC-Q does not account for the operational differences among them. Multimodal traffic should be considered in the future. Given the diversity of potential demand variations, the robust network-level MPC for general demand errors using LTM with queue dynamics requires further investigation. Besides, the impacts of sudden incidents, e.g., traffic accidents or moving bottlenecks, on flow propagation are not considered in the current MPC-Q method. Further research should be carried out to refine the controller design to deal with extreme congestion conditions.

References

[1] K. Aboudolas and N. Geroliminis, "Perimeter and boundary flow control in multi-reservoir heterogeneous networks," *Transp. Res. B, Methodol.*, vol. 55, pp. 265–281, Sep. 2013.

[2] L. Wei, P. Chen, Y. Mei, and Y. Wang, "Turn-level network traffic bottleneck identification using vehicle trajectory data," *Transp. Res. C, Emerg. Technol.*, vol. 140, Jul. 2022, Art. no. 103707.

[3] A. Agarwal, D. Sahu, R. Mohata, K. Jeengar, A. Nautiyal, and D. K. Saxena, "Dynamic traffic signal control for heterogeneous traffic conditions using max pressure and reinforcement learning," *Expert Syst. Appl.*, vol. 254, Nov. 2024, Art. no. 124416.

[4] M. Tajalli, M. Mehrabipour, and A. Hajbabaie, "Network-level coordinated speed optimization and traffic light control for connected and automated vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 11, pp. 6748–6759, Nov. 2021.

[5] O. Mousavizadeh and M. Keyvan-Ekbatani, "On the important features for a well-shaped reduced network MFD estimation during network loading and recovery," *Transp. Res. C, Emerg. Technol.*, vol. 161, Apr. 2024, Art. no. 104539.

[6] C. Diakaki, M. Papageorgiou, and K. Aboudolas, "A multivariable regulator approach to traffic-responsive network-wide signal control," *Control Eng. Pract.*, vol. 10, no. 2, pp. 183–195, Feb. 2002.

[7] K. Aboudolas, M. Papageorgiou, and E. Kosmatopoulos, "Store-and-forward based methods for the signal control problem in large-scale congested urban road networks," *Transp. Res. C, Emerg. Technol.*, vol. 17, no. 2, pp. 163–174, Apr. 2009.

[8] K. Aboudolas, M. Papageorgiou, A. Kouvelas, and E. Kosmatopoulos, "A rolling-horizon quadratic-programming approach to the signal control problem in large-scale congested urban road networks," *Transp. Res. C, Emerg. Technol.*, vol. 18, no. 5, pp. 680–694, Oct. 2010.

[9] F. De Souza, R. C. Carlson, E. R. Müller, and K. Ampountolas, "Multi-commodity traffic signal control and routing with connected vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 5, pp. 4111–4121, May 2022.

[10] Y. Wang, X. Yang, H. Liang, and Y. Liu, "A review of the self-adaptive traffic signal control system based on future traffic environment," *J. Adv. Transp.*, vol. 2018, pp. 1–12, Jun. 2018.

[11] L. Wei, P. Chen, Y. Mei, J. Sun, and Y. Wang, "A hierarchical control framework for alleviating network traffic bottleneck congestion using vehicle trajectory data," *J. Intell. Transp. Syst.*, vol. 28, no. 6, pp. 988–1010, Nov. 2024.

[12] P. Varaiya, "Max pressure control of a network of signalized intersections," *Transp. Res. C, Emerg. Technol.*, vol. 36, pp. 177–195, Nov. 2013.

[13] M. W. Levin, J. Hu, and M. Odell, "Max-pressure signal control with cyclical phase structure," *Transp. Res. C, Emerg. Technol.*, vol. 120, Nov. 2020, Art. no. 102828.

[14] L. Li and S. E. Jabari, "Position weighted backpressure intersection control for urban networks," *Transp. Res. B, Methodol.*, vol. 128, pp. 435–461, Oct. 2019.

[15] J.-J. Henry, J. L. Farges, and J. Tuffal, "The prodyn real time traffic algorithm," in *Control in Transportation Systems*. Amsterdam, The Netherlands: Elsevier, 1984, pp. 305–310.

[16] P. Lowrie, "Scats, sydney co-ordinated adaptive traffic system: A traffic responsive method of controlling urbantraffic," Roads Traffic Authority NSW, Sydney, NSW, Australia, Tech. Rep. 00772163, 1990. [Online]. Available: https://trid.trb.org/view/488852

[17] M. Hajiahmadi, J. Haddad, B. De Schutter, and N. Geroliminis, "Optimal hybrid perimeter and switching plans control for urban traffic networks," *IEEE Trans. Control Syst. Technol.*, vol. 23, no. 2, pp. 464–478, Mar. 2015.

[18] R. Mohebifard and A. Hajbabaie, "Optimal network-level traffic signal control: A benders decomposition-based solution algorithm," *Transp. Res. B, Methodol.*, vol. 121, pp. 252–274, Mar. 2019.

[19] G. S. van de Weg, H. L. Vu, A. Hegyi, and S. P. Hoogendoorn, "A hierarchical control framework for coordination of intersection signal timings in all traffic regimes," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 5, pp. 1815–1827, May 2019.

[20] Z. Zhou, B. De Schutter, S. Lin, and Y. Xi, "Two-level hierarchical model-based predictive control for large-scale urban traffic networks," *IEEE Trans. Control Syst. Technol.*, vol. 25, no. 2, pp. 496–508, Mar. 2017.

[21] J. Zhang, H. Pei, X. J. Ban, and L. Li, "Analysis of cooperative driving strategies at road network level with macroscopic fundamental diagram," *Transp. Res. C, Emerg. Technol.*, vol. 135, Feb. 2022, Art. no. 103503.

[22] M. Saeedmanesh, A. Kouvelas, and N. Geroliminis, "An extended Kalman filter approach for real-time state estimation in multi-region MFD urban networks," *Transp. Res. C, Emerg. Technol.*, vol. 132, Nov. 2021, Art. no. 103384.

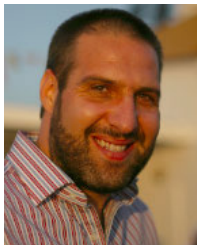
[23] I. I. Sirmatel, D. Tsitsokas, A. Kouvelas, and N. Geroliminis, "Modeling, estimation, and control in large-scale urban road networks with remaining travel distance dynamics," *Transp. Res. C, Emerg. Technol.*, vol. 128, Jul. 2021, Art. no. 103157.

Authorized licensed use limited to: Nanyang Technological University Library. Downloaded on September 10,2026 at 05:12:33 UTC from IEEE Xplore. Restrictions apply.

[24] S. Gao, D. Li, N. Zheng, R. Hu, and Z. She, "Resilient perimeter control for hyper-congested two-region networks with MFD dynamics," *Transp. Res. B, Methodol.*, vol. 156, pp. 50–75, Feb. 2022.
[25] G. S. van de Weg, M. Keyvan-Ekbatani, A. Hegyi, and S. P. Hoogendoorn, "Linear MPC-based urban traffic control using the link transmission model," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 10, pp. 4133–4148, Oct. 2020.
[26] D. Ngoduy, N. H. Hoang, H. L. Vu, and D. Watling, "Optimal queue placement in dynamic system optimum solutions for single origin-destination traffic networks," *Transp. Res. B, Methodol.*, vol. 92, pp. 148–169, Oct. 2016.
[27] N. Raovic, O. A. Nielsen, and C. G. P. Carlo Giacomo Prato, "Dynamic queuing transmission model for dynamic network loading," *Transport*, vol. 32, no. 2, pp. 146–159, Jul. 2015.
[28] L. Wei, S. T. Waller, Y. Mei, P. Chen, Y. Wang, and M. Wang, "A link-based flow model with turn-level queue transmission and time-varying free-flow speed for urban road networks," *Transp. Res. C, Emerg. Technol.*, vol. 170, Jan. 2025, Art. no. 104937.
[29] S. Hu et al., "High time-resolution queue profile estimation at signalized intersections based on extended Kalman filtering," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 21274–21290, Nov. 2022.
[30] G. Zhang, F. Chang, J. Jin, F. Yang, and H. Huang, "Multi-objective deep reinforcement learning approach for adaptive traffic signal control system with concurrent optimization of safety, efficiency, and decarbonization at intersections," *Accident Anal. Prevention*, vol. 199, May 2024, Art. no. 107451.
[31] T. Wang, Z. Zhu, J. Zhang, J. Tian, and W. Zhang, "A large-scale traffic signal control algorithm based on multi-layer graph deep reinforcement learning," *Transp. Res. C, Emerg. Technol.*, vol. 162, May 2024, Art. no. 104582.
[32] R. Zhou, T. Nousch, L. Wei, and M. Wang, "Constrained traffic signal control under competing public transport priority requests via safe reinforcement learning," *Expert Syst. Appl.*, vol. 284, Jul. 2025, Art. no. 127676.
[33] B. Zhou, Q. Zhou, S. Hu, D. Ma, S. Jin, and D.-H. Lee, "Cooperative traffic signal control using a distributed agent-based deep reinforcement learning with incentive communication," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 8, pp. 10147–10160, Aug. 2024.
[34] F. Ren, W. Dong, X. Zhao, F. Zhang, Y. Kong, and Q. Yang, "Two-layer coordinated reinforcement learning for traffic signal control in traffic network," *Expert Syst. Appl.*, vol. 235, Jan. 2024, Art. no. 121111.
[35] X. Sun and Y. Yin, "A simulation study on max pressure control of signalized intersections," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 2672, no. 18, pp. 117–127, 2018.



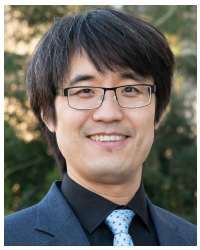
Lei Wei received the Ph.D. degree from Beihang University, China, in 2022. He is currently a Post-Doctoral Researcher with the "Friedrich List" Faculty of Transport and Traffic Sciences, TU Dresden, Germany. His research interests include urban traffic operation and control and macroscopic traffic flow modeling and simulation.



Konstantinos Ampountolas (Member, IEEE) received the Dipl.Ing. degree in production engineering and management, the M.Sc. degree in operations research, and the Ph.D. degree in engineering from the Technical University of Crete, Greece, in 1999, 2002, and 2009, respectively. He was a Post-Doctoral Researcher with the Centre for Research and Technology Hellas, Greece, in 2010, a Visiting Researcher Scholar with the University of California at Berkeley, Berkeley, CA, USA, in 2011, a Senior Lecturer with the James Watt School of Engineering, University of Glasgow, U.K., from 2013 to 2019, and a Research Fellow with EPFL, Switzerland, from 2012 to 2013.. Since 2019, he has been an Associate Professor with the Department of Mechanical Engineering, University of Thessaly, Greece. His research interests include automatic control and optimization and their applications to transport systems. He served as an Editor for *Transportation* and *Data in Brief* (2018–2019), an Associate Editor for *Data Science for Transportation* (2018–2020), and on the Editorial Advisory Board for *Transportation Research Part—C* (2014–2021) and *Transportation Research Procedia* (since 2014).



Angelika Hirrle received the Ph.D. degree from TU Dresden in 2017. She is currently a Research Associate with the "Friedrich List" Faculty of Transport and Traffic Sciences, TU Dresden, Germany. Her research interests include sustainable multi-modal urban transport, in particular modeling, optimization, and control aspects.



Meng Wang (Senior Member, IEEE) received the Ph.D. degree (Hons.) from TU Delft in 2014. He was a Post-Doctoral Researcher with the Faculty of Mechanical Engineering, TU Delft, from 2014 to 2015, and an Assistant Professor with the Department of Transport and Planning, TU Delft, from 2015 to 2021. Since 2021, he has been a Full Professor and the Head of the Chair of Traffic Process Automation, TU Dresden. His research interests include traffic flow modeling and control, control design, and impact assessment of connected and automated transport systems. He is an Associate Editor of IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS and *Transportmetrica B*.

Authorized licensed use limited to: Nanyang Technological University Library. Downloaded on September 10,2026 at 05:12:33 UTC from IEEE Xplore. Restrictions apply.